

Microwave Transmission Line Characterization Using a Handheld VNA

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1 Objectives

The objective of this experiment is to determine whether the provided 50-ohm coaxial transmission line behaves as a distortionless transmission line within the operating frequency range of 1–5 GHz. Using measured S-parameters, the attenuation constant (α) and phase constant (β) are extracted and analyzed.

2 Equipment Used

- NanoVNA-F V3 Handheld Vector Network Analyzer
- Two coaxial cables (one for measurement, one for through calibration)
- Calibration kit (Open, Short, Load, Through)
- Computer with MATLAB installed

3 Theory

The propagation constant of a microwave transmission line is defined as:

$$\gamma = \alpha + j\beta$$

The complex propagation function is:

$$X = e^{-\gamma l} = e^{-\alpha l} e^{-j\beta l}$$

The intermediate parameter Q is computed from the measured S-parameters as:

$$Q = \frac{S_{11}^2 + S_{21}^2 + 1}{2S_{11}}$$

The reflection coefficient is:

$$\Gamma = Q \pm \sqrt{Q^2 - 1}$$

Finally, the propagation function is:

$$X = \frac{(S_{11} + S_{21}) - \Gamma}{1 - (S_{11} + S_{21})}$$

From X , the attenuation and phase constants are extracted as:

$$\alpha = -\frac{\ln |X|}{l}, \quad \beta = -\frac{\arg(X)}{l}$$

4 Measurements

Measured magnitude and phase of S_{11} and S_{21} at 1–5 GHz.

Parameter	1 GHz	2 GHz	3 GHz	4 GHz	5 GHz
S_{11} (dB)	-32	-32.65	-29.68	-20.90	-8.63
S_{11} Phase (deg)	-66.1	-108.74	157.05	155.34	65.99
S_{21} (dB)	2.96	3.26	1.63	3.88	1.09
S_{21} Phase (deg)	-52.32	-95.59	-120.71	175.78	134.32

5 Measurement Pictures

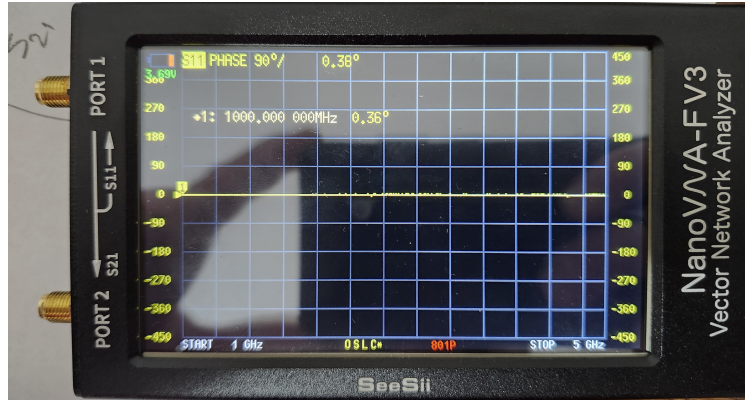


Figure 1: Calibration completed.

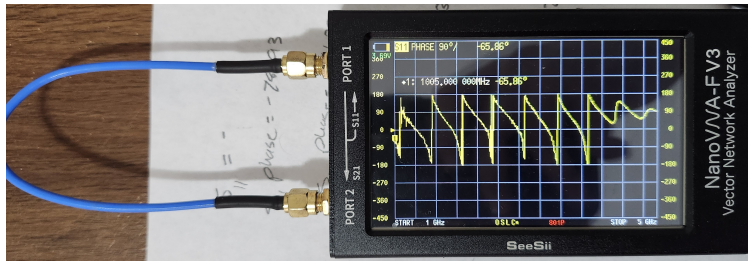


Figure 2: Cable connected to VNA..

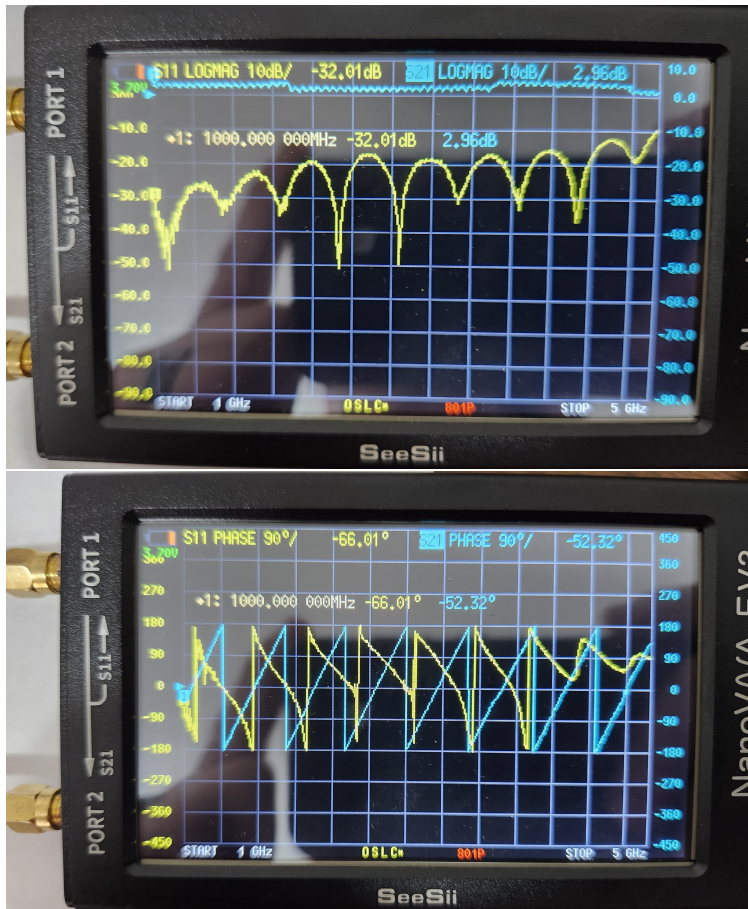


Figure 3: S-parameter measurement at 1 GHz.

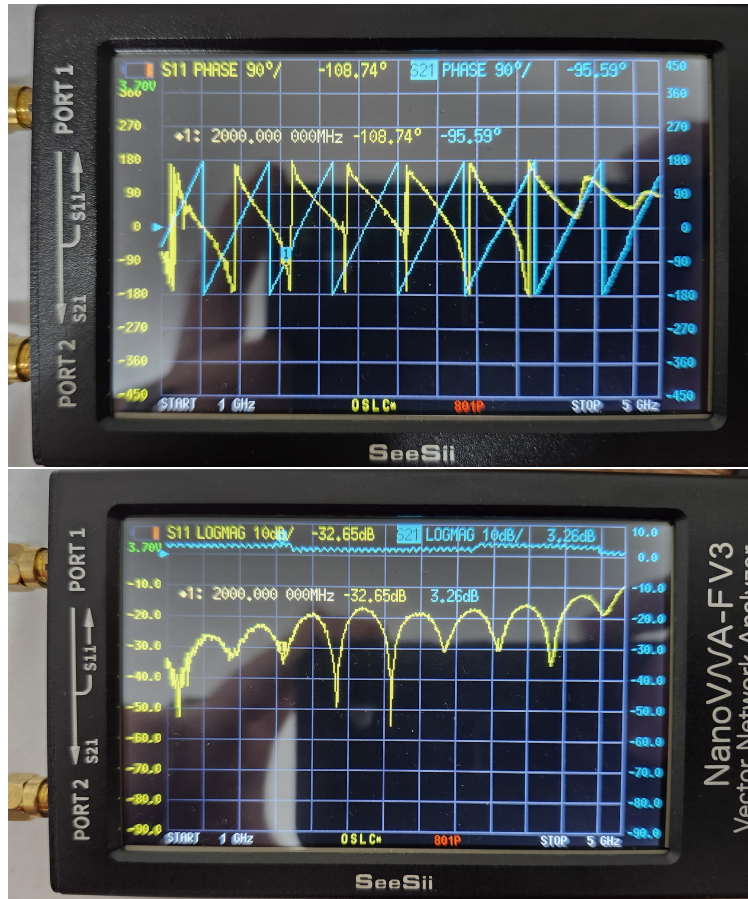


Figure 4: S-parameter measurement at 2 GHz.

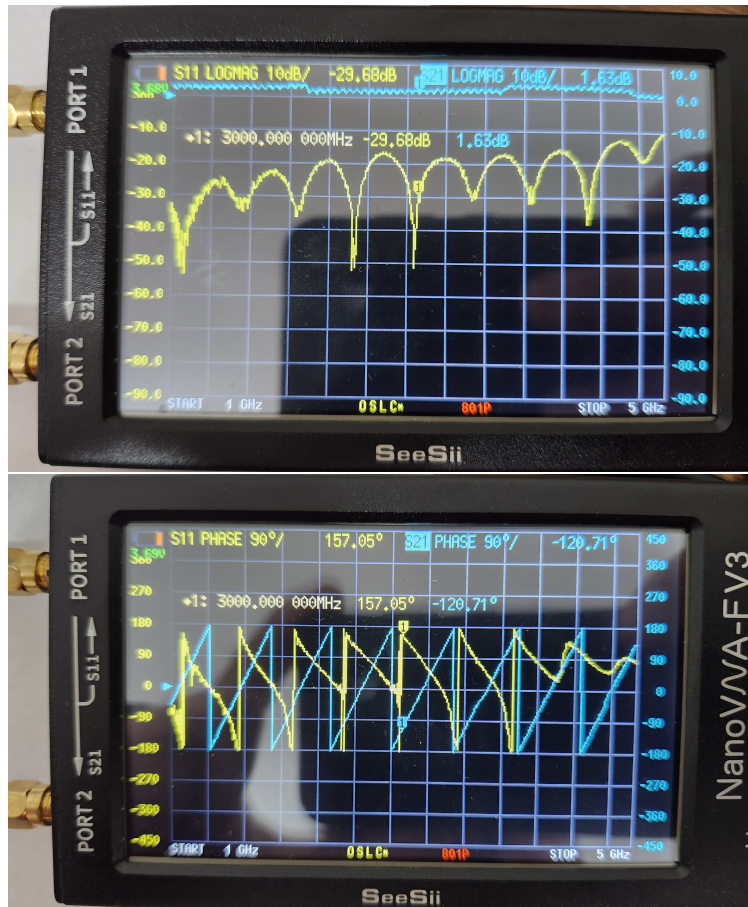


Figure 5: S-parameter measurement at 3 GHz.

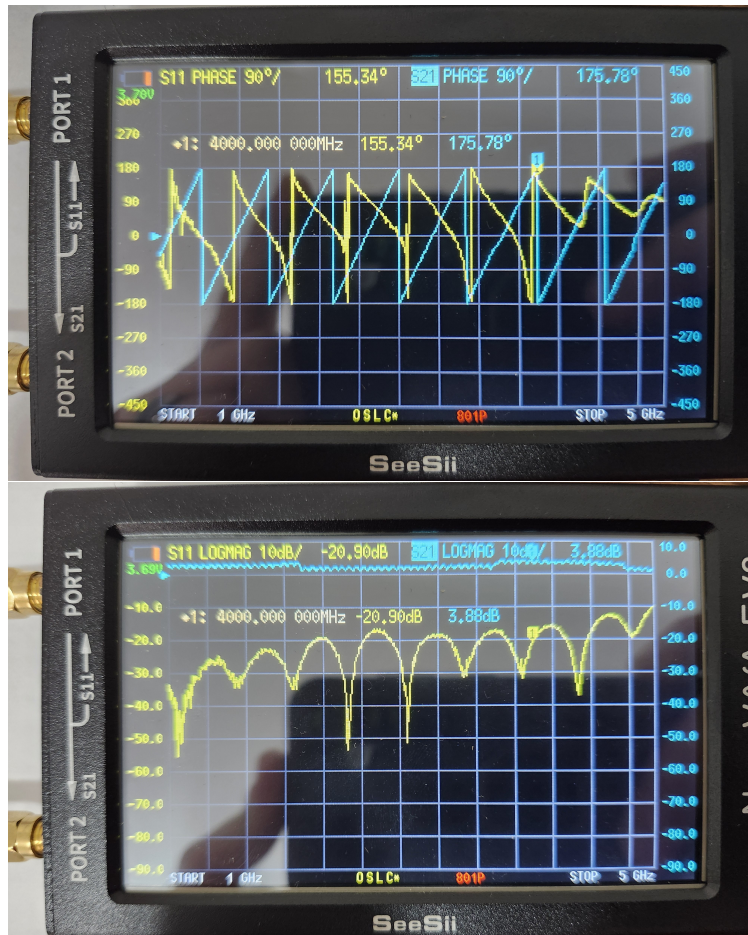


Figure 6: S-parameter measurement at 4 GHz.

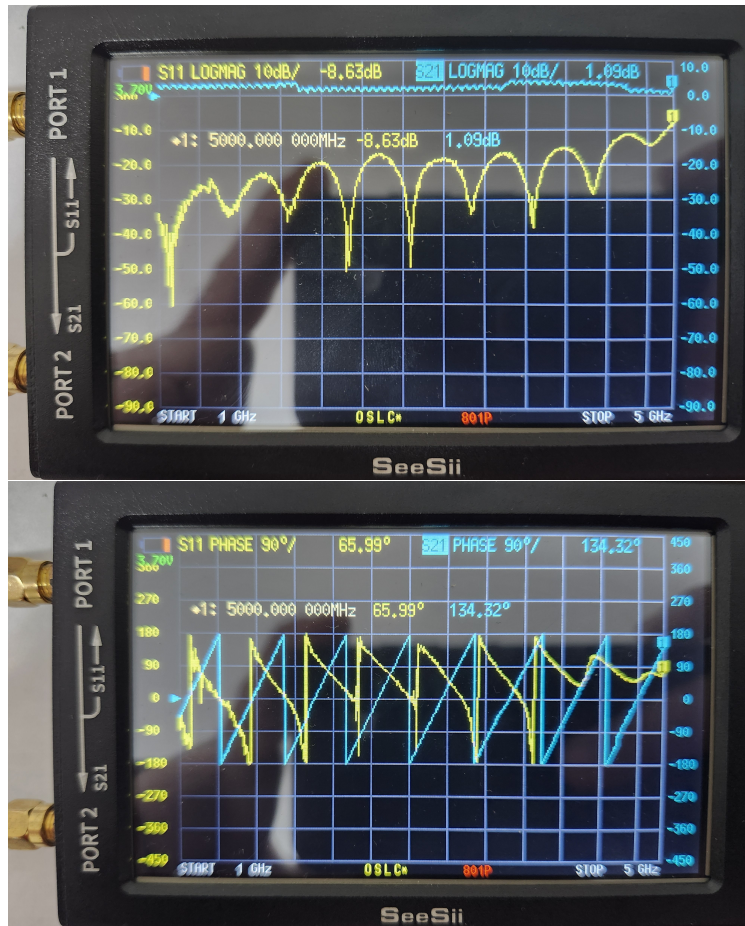


Figure 7: S-parameter measurement at 5 GHz.

6 MATLAB Code

```
% Frequency (Hz)
f_GHz = [1 2 3 4 5]*1e9;

% Cable length (meter)
l = 0.20;

% S11 Values
S11_dB = [-32 -32.65 -29.68 -20.90 -8.63];
S11_phase_deg = [-66.1 -108.74 157.05 155.34 65.99];

% Convert S11 to complex numbers
S11_mag = 10.^(S11_dB/20);
S11_phase = deg2rad(S11_phase_deg);
S11 = S11_mag .* exp(1j*S11_phase);

% S21 Values
```

```

S21_dB = [2.96 3.26 1.63 3.88 1.09];
S21_phase_deg = [-52.32 -95.59 -120.71 175.78 134.32];

% Convert S21 to complex numbers
S21_mag = 10.^(S21_dB/20);
S21_phase = deg2rad(S21_phase_deg);
S21 = S21_mag .* exp(1j*S21_phase);

% Calculate Q
Q = (S11.^2 + S21.^2 + 1) ./ (2*S11);

% Compute Gamma
Gamma1 = Q + sqrt(Q.^2 - 1);
Gamma2 = Q - sqrt(Q.^2 - 1);
Gamma = Gamma1;
Gamma(abs(Gamma1) > 1) = Gamma2(abs(Gamma1) > 1);

% Compute X
X = (((S11 + S21) - Gamma) ./ (1 - (S11 + S21)));

% Compute alpha and beta
alpha = -log(abs(X)) ./ 1;
beta = -angle(X) ./ 1;

% Plotting
subplot(1,2,1);
plot(f_GHz, alpha, 'LineWidth', 2);
xlabel('Frequency (GHz)'); ylabel('\alpha (Np/m)');
title('Attenuation Constant \alpha vs Frequency'); grid on;

subplot(1,2,2);
plot(f_GHz, beta, 'LineWidth', 2);
xlabel('Frequency (GHz)'); ylabel('\beta (rad/m)');
title('Phase Constant \beta vs Frequency'); grid on;

```

FINAL RESULTS

ans =

5×6 [table](#)

Freq_Hz	Q	Gamma	X	Alpha	Beta
1	38.85-6.3318i	0.012539+0.0020443i	-0.90327-0.85832i	-1.0999	11.909
2	-0.91703-24.744i	-0.00074692+0.020171i	-0.67835-0.42913i	1.0989	12.888
3	3.3175-19.734i	0.0041347+0.024626i	-0.57065-0.28212i	2.2582	13.412
4	-18.093-6.0803i	-0.024841+0.0083597i	-0.61304+0.019201i	2.4442	-15.551
5	-0.97701-1.7329i	-0.10875+0.21704i	-0.48462+0.22864i	3.1195	-13.504

Figure 8: Final results.

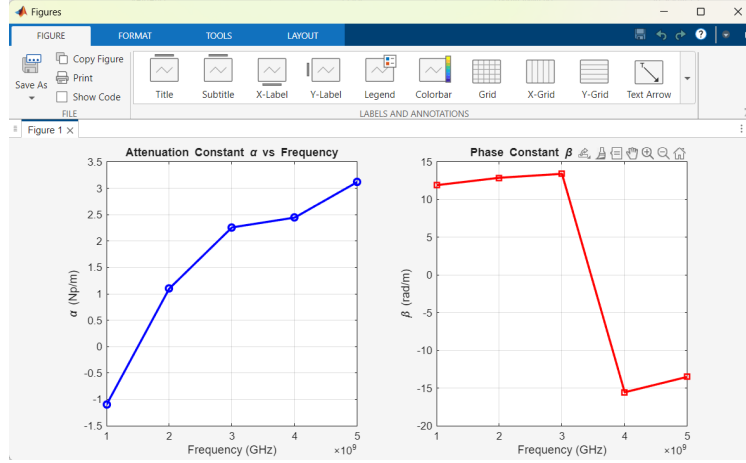


Figure 9: Extracted attenuation constant (α) and phase constant β vs frequency.

7 Observation

The S-parameter measurements indicate increasing attenuation with frequency, consistent with expected coaxial cable behavior. The extracted α values rise at higher frequencies, confirming higher losses. The β variation remains approximately proportional to frequency, showing that phase response is mostly linear with small fluctuations due to measurement noise and VNA resolution limits.

8 Discussion

Based on the extracted attenuation and phase constants, the coaxial cable does not behave as a perfectly distortionless transmission line. For a transmission line to be distortionless, α must remain constant with frequency, and β must vary linearly. Our results show increasing attenuation and minor deviations in phase linearity. As an engineer, the cable may be considered approximately distortionless only within the lower-frequency region (1–3 GHz), where losses remain small and phase behavior is nearly ideal. Beyond this range, distortion becomes increasingly significant.